



RAMCO INSTITUTE OF TECHNOLOGY

Approved by AICTE, New Delhi & Affiliated to Anna University
Accredited by NAAC & An ISO 9001:2015 Certified Institution
NBA Accredited UG Programs: CSE, EEE, ECE and MECH

Department of Computer Science and Engineering Regulation-2021 Course Outcomes

Semester 1	
HS3152 PROFESSIONAL ENGLISH I	CO1 To use appropriate words in a professional context CO2 To gain understanding of basic grammatic structures and use them in right context CO3 To read and infer the denotative and connotative meanings of technical texts CO4 To write definitions, descriptions, narrations and essays on various topics
MA3151 MATRICES AND CALCULUS	CO1 Use the matrix algebra methods for solving practical problems. CO2 Apply differential calculus tools in solving various application problems. CO3 Able to use differential calculus ideas on several variable functions. CO4 Apply different methods of integration in solving practical problems. CO5 Apply multiple integral ideas in solving areas, volumes and other practical problems.
PH3151 ENGINEERING PHYSICS	CO1 Understand the importance of mechanics. CO2 Express their knowledge in electromagnetic waves. CO3 Demonstrate a strong foundational knowledge in oscillations, optics and lasers. CO4 Understand the importance of quantum physics. CO5 Comprehend and apply quantum mechanical principles towards the formation of energy bands.
CY3151 ENGINEERING CHEMISTRY	CO1 To infer the quality of water from quality parameter data and propose suitable treatment methodologies to treat water. CO2 To identify and apply basic concepts of nanoscience and nanotechnology in designing the synthesis of nanomaterials for engineering and technology applications. CO3 To apply the knowledge of phase rule and composites for material selection requirements. CO4 To recommend suitable fuels for engineering processes and applications. CO5 To recognize different forms of energy resources and apply them for suitable applications in energy sectors.

GE3151 PROBLEM SOLVING AND PYTHON PROGRAMMING	CO1 Develop algorithmic solutions to simple computational problems. CO2 Develop and execute simple Python programs. CO3 Write simple Python programs using conditionals and loops for solving problems. CO4 Decompose a Python program into functions. CO5 Represent compound data using Python lists, tuples, dictionaries etc.
GE3171 PROBLEM SOLVING AND PYTHON PROGRAMMING LABORATORY	CO1 Develop algorithmic solutions to simple computational problems CO2 Develop and execute simple Python programs. CO3 Implement programs in Python using conditionals and loops for solving problems. CO4 Deploy functions to decompose a Python program. CO5 Process compound data using Python data structures. CO6 Utilize Python packages in developing software applications.
BS3171 PHYSICS AND CHEMISTRY LABORATORY	CO1 Understand the functioning of various physics laboratory equipment. CO2 Use graphical models to analyze laboratory data. CO3 Use mathematical models as a medium for quantitative reasoning and describing physical reality. CO4 Access, process and analyze scientific information. CO5 Solve problems individually and collaboratively. CO6 To analyse the quality of water samples with respect to their acidity, alkalinity, hardness and DO. CO7 To determine the amount of metal ions through volumetric and spectroscopic techniques CO8 To analyse and determine the composition of alloys. CO9 To learn simple method of synthesis of nanoparticles CO10 To quantitatively analyse the impurities in solution by electroanalytical techniques
GE3172 ENGLISH LABORATORY	CO1 To listen to and comprehend general as well as complex academic information CO2 To listen to and understand different points of view in a discussion CO3 To speak fluently and accurately in formal and informal communicative contexts CO4 To describe products and processes and explain their uses and purposes clearly and accurately CO5 To express their opinions effectively in both formal and informal discussions

Semester 2	
HS3252 PROFESSIONAL ENGLISH - II	CO1 To compare and contrast products and ideas in technical texts. CO2 To identify and report cause and effects in events, industrial processes through technical texts CO3 To analyse problems in order to arrive at feasible solutions and communicate them in the written format. CO4 To present their ideas and opinions in a planned and logical manner CO5 To draft effective resumes in the context of job search.
MA3251 STATISTICS AND NUMERICAL METHODS	CO1 Apply the concept of testing of hypothesis for small and large samples in real life problems. CO2 Apply the basic concepts of classifications of design of experiments in the field of agriculture. CO3 Appreciate the numerical techniques of interpolation in various intervals and apply the numerical techniques of differentiation and integration for engineering problems. CO4 Understand the knowledge of various techniques and methods for solving first and second order ordinary differential equations. CO5 Solve the partial and ordinary differential equations with initial and boundary conditions by using certain techniques with engineering applications.
PH3256 PHYSICS FOR INFORMATION SCIENCE	CO1 Gain knowledge on classical and quantum electron theories, and energy band structures CO2 Acquire knowledge on basics of semiconductor physics and its applications in various devices CO3 Get knowledge on magnetic properties of materials and their applications in data storage, CO4 Have the necessary understanding on the functioning of optical materials for optoelectronics CO5 Understand the basics of quantum structures and their applications and basics of quantum
BE3251 BASIC ELECTRICAL AND ELECTRONICS ENGINEERING	CO1 Use BIS conventions and specifications for engineering drawing. CO2 Construct the conic curves, involutes and cycloid. CO3 Solve practical problems involving projection of lines. CO4 Draw the orthographic, isometric and perspective projections of simple solids. CO5 Draw the development of simple solids.

GE3251 ENGINEERING GRAPHICS	CO1 Summarize the importance of environment, biodiversity, ecosystem and how to solve environmental related problems. CO2 Describe the causes, effect and control measures of air pollution, water pollution, soil pollution, noise pollution, radioactive pollution and thermal pollution with their relevant case studies. CO3 Discuss the various renewable and non-renewable resources and energy conservation processes. CO4 Explain the social issues and solutions for sustainable environment with relevant Acts and case studies. CO5 Review the impact of human population in the environment and its remedial measures.
CS3251 PROGRAMMING IN C	CO1 Demonstrate knowledge on C Programming constructs CO2 Develop simple applications in C using basic constructs CO3 Design and implement applications using arrays and strings CO4 Develop and implement modular applications in C using functions. CO5 Develop applications in C using structures and pointers.
GE3271 ENGINEERING PRACTICES LABORATORY	CO1 Draw pipe line plan; lay and connect various pipe fittings used in common household plumbing work; Saw; plan; make joints in wood materials used in common household wood work. CO2 Wire various electrical joints in common household electrical wire work. Weld various joints in steel plates using arc welding work; Machine various simple processes like turning, drilling, tapping in parts; Assemble simple mechanical assembly of common household equipments; Make a tray out of metal sheet using sheet metal work. CO3 CO4 Solder and test simple electronic circuits; Assemble and test simple electronic components on PCB
CS3271 PROGRAMMING IN C LABORATORY	CO1 Demonstrate knowledge on C programming constructs. CO2 Develop programs in C using basic constructs. CO3 Develop programs in C using arrays. CO4 Develop applications in C using strings, pointers, functions. CO5 Develop applications in C using structures.
GE3272 COMMUNICATION LABORATORY	CO1 Speak effectively in group discussions held in a formal/semi formal contexts. CO2 Discuss, analyse and present concepts and problems from various perspectives to arrive at suitable solutions CO3 Write emails, letters and effective job applications. CO4 Write critical reports to convey data and information with clarity and precision CO5 Give appropriate instructions and recommendations for safe execution of tasks

Semester 3	
MA3354 DISCRETE MATHEMATICS	CO1 Have knowledge of the concepts needed to test the logic of a program. CO2 Have an understanding in identifying structures on many levels. Be aware of a class of functions which transform a finite set into another finite set which relates to input and output functions in computer science. CO3 Be aware of the counting principles. CO4 Be exposed to concepts and properties of algebraic structures such as groups, rings and fields. CO5
CS3351 DIGITAL PRINCIPLES AND COMPUTER ORGANIZATION	CO1 Design various combinational digital circuits using logic gates CO2 Design sequential circuits and analyze the design procedures CO3 State the fundamentals of computer systems and analyze the execution of an instruction CO4 Analyze different types of control design and identify hazards CO5 Identify the characteristics of various memory systems and I/O communication
CS3352 FOUNDATIONS OF DATA SCIENCE	CO1 Define the data science process CO2 Understand different types of data description for data science process CO3 Gain knowledge on relationships between data CO4 Use the Python Libraries for Data Wrangling CO5 Apply visualization Libraries in Python to interpret and explore data
CS3301 DATA STRUCTURES	CO1 Define linear and non-linear data structures. CO2 Implement linear and non-linear data structure operations. CO3 Use appropriate linear/non-linear data structure operations for solving a given problem. CO4 Apply appropriate graph algorithms for graph applications. CO5 Analyze the various searching and sorting algorithms.
CS3391 OBJECT ORIENTED PROGRAMMING	CO1 Apply the concepts of classes and objects to solve simple problems CO2 Develop programs using inheritance, packages and interfaces CO3 Make use of exception handling mechanisms and multithreaded model to solve real world problems CO4 Build Java applications with I/O packages, string classes, Collections and generics concepts CO5 Integrate the concepts of event handling and JavaFX components and controls for developing GUI based applications
CS3311 DATA STRUCTURES LABORATORY	CO1 Implement Linear data structure algorithms. CO2 Implement applications using Stacks and Linked lists CO3 Implement Binary Search tree and AVL tree operations. CO4 Implement graph algorithms. CO5 Analyze the various searching and sorting algorithms
	CO1 Design and develop java programs using object oriented programming concepts

CS3381 OBJECT ORIENTED PROGRAMMING LABORATORY	CO2 Develop simple applications using object oriented concepts such as package, exceptions CO3 Implement multithreading, and generics concepts CO4 Create GUIs and event driven programming applications for real world problems CO5 Implement and deploy web applications using Java
CS3361 DATA SCIENCE LABORATORY	CO1 Make use of the python libraries for data science CO2 Make use of the basic Statistical and Probability measures for data science. CO3 Perform descriptive analytics on the benchmark data sets. CO4 Perform correlation and regression analytics on standard data sets CO5 Present and interpret data using visualization packages in Python.
GE3361 PROFESSIONAL DEVELOPMENT	CO1 Use MS Word to create quality documents, by structuring and organizing content for their day to day technical and academic requirements CO2 Use MS EXCEL to perform data operations and analytics, record, retrieve data as per requirements and visualize data for ease of understanding CO3 Use MS PowerPoint to create high quality academic presentations by including common tables, charts, graphs, interlinking other elements, and using media objects.
Semester 4	
CS3452 THEORY OF COMPUTATION	CO1 Construct automata theory using Finite Automata CO2 Write regular expressions for any pattern CO3 Design context free grammar and Pushdown Automata CO4 Design Turing machine for computational functions CO5 Differentiate between decidable and undecidable problems
CS3491 ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING	CO1 Use appropriate search algorithms for problem solving CO2 Apply reasoning under uncertainty CO3 Build supervised learning models CO4 Build ensembling and unsupervised models CO5 Build deep learning neural network models
CS3492 DATABASE MANAGEMENT SYSTEMS	CO1 Construct SQL Queries using relational algebra CO2 Design database using ER model and normalize the database CO3 Construct queries to handle transaction processing and maintain consistency of the database CO4 Compare and contrast various indexing strategies and apply the knowledge to tune the performance of the database CO5 Appraise how advanced databases differ from Relational Databases and find a suitable database for the given requirement.
CS3401 ALGORITHMS	CO1 Analyze the efficiency of algorithms using various frameworks CO2 Apply graph algorithms to solve problems and analyze their efficiency.

	CO3 Make use of algorithm design techniques like divide and conquer, dynamic programming and greedy techniques to solve problems CO4 Use the state space tree method for solving problems. CO5 Solve problems using approximation algorithms and randomized algorithms
CS3451 INTRODUCTION TO OPERATING SYSTEMS	CO1 Analyze various scheduling algorithms and process synchronization. CO2 Explain deadlock prevention and avoidance algorithms. CO3 Compare and contrast various memory management schemes. CO4 Explain the functionality of file systems, I/O systems, and Virtualization CO5 Compare iOS and Android Operating Systems.
GE3451 ENVIRONMENTAL SCIENCES AND SUSTAINABILITY	CO1 To recognize and understand the functions of environment, ecosystems and biodiversity and their conservation. CO2 To identify the causes, effects of environmental pollution and natural disasters and contribute to the preventive measures in the society. CO3 To identify and apply the understanding of renewable and non-renewable resources and contribute to the sustainable measures to preserve them for future generations. CO4 To recognize the different goals of sustainable development and apply them for suitable technological advancement and societal development. CO5 To demonstrate the knowledge of sustainability practices and identify green materials, energy cycles and the role of sustainable urbanization.
CS3461 OPERATING SYSTEMS LABORATORY	CO1 Define and implement UNIX Commands. CO2 Compare the performance of various CPU Scheduling Algorithms. CO3 Compare and contrast various Memory Allocation Methods. CO4 Define File Organization and File Allocation Strategies. CO5 Implement various Disk Scheduling Algorithms
CS3481 DATABASE MANAGEMENT SYSTEMS LABORATORY	CO1 Create databases with different types of key constraints. CO2 Construct simple and complex SQL queries using DML and DCL commands. CO3 Use advanced features such as stored procedures and triggers and incorporate in GUI based application development. CO4 Create an XML database and validate with meta-data (XML schema). CO5 Create and manipulate data using NOSQL database.
Semester 5	
CS3591 COMPUTER NETWORKS	CO1 Explain the basic layers and its functions in computer networks. CO2 Understand the basics of how data flows from one node to another. CO3 Analyze routing algorithms. CO4 Describe protocols for various functions in the network.

	CO5	Analyze the working of various application layer protocols.
CS3501 COMPILER DESIGN	CO1	Understand the techniques in different phases of a compiler.
	CO2	Design a lexical analyser for a sample language and learn to use the LEX tool.
	CO3	Apply different parsing algorithms to develop a parser and learn to use YACC tool
	CO4	Understand semantics rules (SDT), intermediate code generation and run-time environment.
	CO5	Implement code generation and apply code optimization techniques.
CB3491 CRYPTOGRAPHY AND CYBER SECURITY	CO1	Understand the fundamentals of networks security, security architecture, threats and vulnerabilities
	CO2	Apply the different cryptographic operations of symmetric cryptographic algorithms
	CO3	Apply the different cryptographic operations of public key cryptography
	CO4	Apply the various Authentication schemes to simulate different applications
	CO5	Understand various cyber crimes and cyber security.
CS3551 DISTRIBUTED COMPUTING	CO1	Explain the foundations of distributed systems
	CO2	Solve synchronization and state consistency problems
	CO3	Use resource sharing techniques in distributed systems
	CO4	Apply working model of consensus and reliability of distributed systems
	CO5	Explain the fundamentals of cloud computing
Semester 6		
CCS356 OBJECT ORIENTED SOFTWARE ENGINEERING	CO1	Compare various Software Development Lifecycle Models
	CO2	Evaluate project management approaches as well as cost and schedule estimation strategies.
	CO3	Perform formal analysis on specifications.
	CO4	Use UML diagrams for analysis and design.
	CO5	Architect and design using architectural styles and design patterns, and test the system
CS3691 EMBEDDED SYSTEMS AND IOT	CO1	Explain the architecture of embedded processors.
	CO2	Write embedded C programs.
	CO3	Design simple embedded applications.
	CO4	Compare the communication models in IOT
	CO5	Design IoT applications using Arduino/Raspberry Pi /open platform.
Semester 7		
CS3711 SUMMER INTERNSHIP	CO1	Industry Practices, Processes, Techniques, technology, automation and other core aspects of software industry
	CO2	Analyze, Design solutions to complex business problems
	CO3	Build and deploy solutions for target platform
	CO4	Preparation of Technical reports and presentation.

Semester 8		
CS3811 PROJECT WORK/ INTERNSHIP	CO1 Gain Domain knowledge and technical skill set required for solving industry / research problems CO2 Provide solution architecture, module level designs, algorithms CO3 Implement, test and deploy the solution for the target platform CO4 Prepare detailed technical report, demonstrate and present the work	
Electives		
CCS346 EXPLORATORY DATA ANALYSIS	CO1 Understand the fundamentals of exploratory data analysis. CO2 Implement the data visualization using Matplotlib. CO3 Perform univariate data exploration and analysis. CO4 Apply bivariate data exploration and analysis. CO5 Use Data exploration and visualization techniques for multivariate and time series data.	
CCS334 BIG DATA ANALYTICS	CO1 Describe big data and use cases from selected business domains. CO2 Explain NoSQL big data management. CO3 Install, configure, and run Hadoop and HDFS. CO4 Perform map-reduce analytics using Hadoop. CO5 Use Hadoop-related tools such as HBase, Cassandra, Pig, and Hive for big data analytics.	
CCS375 WEB TECHNOLOGIES	CO1 Construct a basic website using HTML and Cascading Style Sheets CO2 Build dynamic web page with validation using Java Script objects and by applying different event handling mechanisms. CO3 Develop server side programs using Servlets and JSP. CO4 Construct simple web pages in PHP and to represent data in XML format. CO5 Develop interactive web applications.	
CCS370 UI AND UX DESIGN	CO1 Build UI for user Applications CO2 Evaluate UX design of any product or application CO3 Demonstrate UX Skills in product development CO4 Implement Sketching principles CO5 Create Wireframe and Prototype	
CCS344 ETHICAL HACKING	CO1 To express knowledge on basics of computer based vulnerabilities CO2 To gain understanding on different foot printing, reconnaissance and scanning methods. CO3 To demonstrate the enumeration and vulnerability analysis methods CO4 To gain knowledge on hacking options available in Web and wireless applications. CO5 To acquire knowledge on the options for network protection. CO6 To use tools to perform ethical hacking to expose the vulnerabilities.	
	CO1 Have knowledge on digital forensics.	

CCS343 DIGITAL AND MOBILE FORENSICS	CO2 Know about digital crime and investigations. CO3 Be forensic ready. CO4 Investigate, identify and extract digital evidence from iOS devices. CO5 Investigate, identify and extract digital evidence from Android devices.
CCS339 CRYPTOCURRENCY AND BLOCKCHAIN TECHNOLOGIES	CO1 Understand emerging abstract models for Blockchain Technology CO2 Identify major research challenges and technical gaps existing between theory and practice in the crypto currency domain. CO3 It provides conceptual understanding of the function of Blockchain as a method of securing distributed ledgers, how consensus on their contents is achieved, and the new applications that they enable. CO4 Apply hyperledger Fabric and Ethereum platform to implement the Block chain Application.
CCW332 DIGITAL MARKETING	CO1 To examine and explore the role and importance of digital marketing in today's rapidly changing business environment.. CO2 To focuses on how digital marketing can be utilized by organizations and how its effectiveness can be measured. CO3 To know the key elements of a digital marketing strategy. CO4 To study how the effectiveness of a digital marketing campaign can be measured CO5 To demonstrate advanced practical skills in common digital marketing tools such as SEO, SEM, Social media and Blogs.